8 **(a)** W m^{-1} K^{-1} [1]

(b) Rate of heat flow = power carried by water [1]

= (m / t)(c)(32.5 − 30)

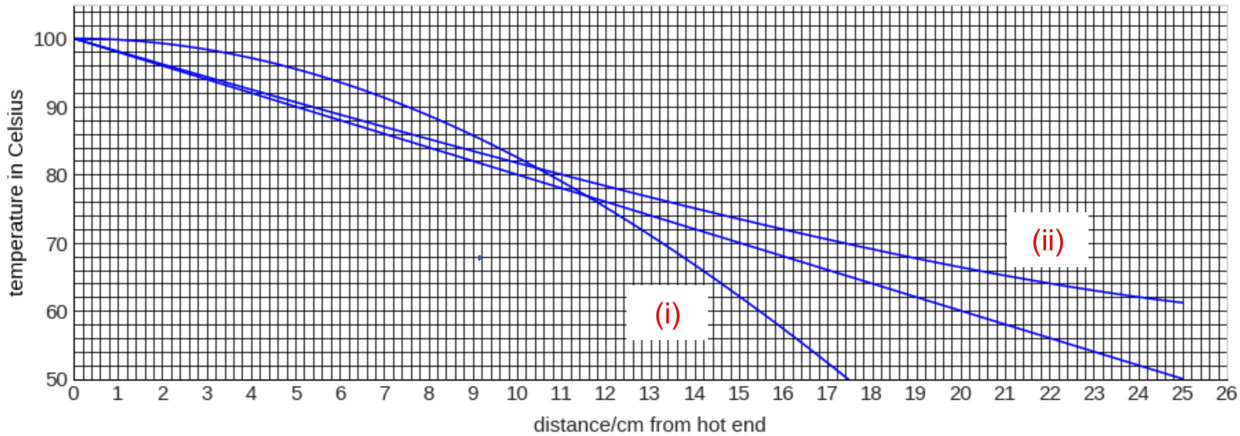
= 32.6 W [1]

(c) 32.6 = (k)(π)$(1.5 \times 10^{-2})^2$(gradient) [1, ecf]

32.6 = (k)(π)$(1.5 \times 10^{-2})^2$ [(100 – 50) / 0.25] [1]

k = 230 [1]

(d)



(i) *A* decrease with distance, so (magnitude of) *T*-*x* gradient increases with distance. [correct line, 1]

(ii) Without insulation, dQ/dt decreases along the length, therefore gradient decreases with distance.

[explanation, 1]

[correct line, 1]

(iii) There will be a sharp drop in temperature over a small distance. All the temperature values will be the same beyond a small distance [1]

Use a short length (or small d) of material instead (or large A) [1]

(e) **(i)** Total area of container = $2\pi\left(10\times10^{-2}\right)^2 + 2\pi\left(10\times10^{-2}\right)\left(16\times10^{-2}\right) = 0.163\ m^2$ [1]

Heat conducted into container 120 s = $(2.50)(0.163)\left(\frac{30}{2\times10^{-2}}\right)(120)$ = 73476 J

[1,ecf]

Mass melted away = $\frac{73476}{3.35\times10^{5}}$ = 0.219 kg [1,ecf]

Assumption: heat conducted through wall of container only. [1]

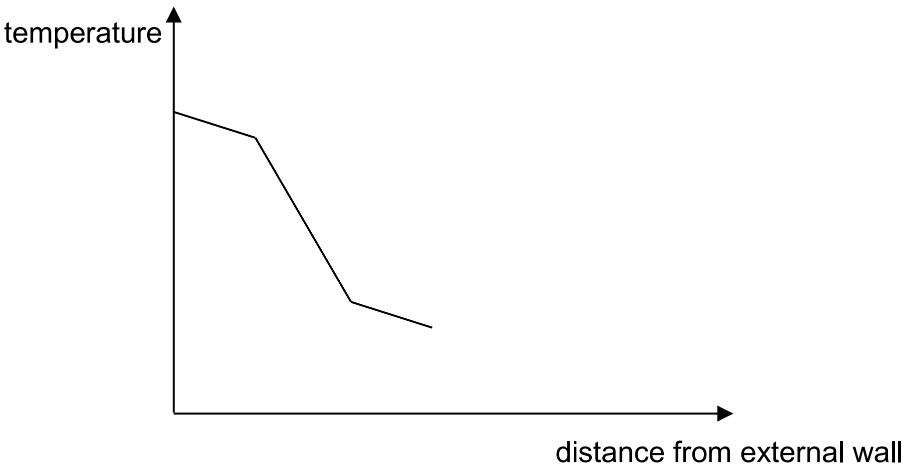
(ii) Rate of heat absorbed by radiation

= $(5.7 \times 10^{-8})(0.1632)(303.15^4 - 273.15^4)$ [1]

= 26.8 J [1]

Compared to conduction, the rate of heat loss by radiation is small (< 5%) [1]

(iii) The gradient is steeper in the air gap [1]
The two insulating layer has the same gradient. [1]



(iv) Thicker layer of air allows air to flow or circulate when it is further away from the wall. [1]

Convection current circulation can happen with a thick layer of air, helping the transfer heat energy. [1]